

Weisi Chen

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Brief Biography

I hold a postgraduate qualification in Health Data Science (WAM: 88%). Since 2023, I have contributed to research across the fields of biostatistics, clinical epidemiology, and pharmacoepidemiology.

Over the years, I have worked extensively with large, linked administrative health datasets, developing strong expertise in data integration, data quality assurance, and population-level analysis. I have experience working with the SURE, PLIDA, and NHDH data platforms and am proficient in statistical software, including SAS and R.

Technical Skills & Expertise

SAS · R · R Shiny · GitHub

Big Data Management · Reproducible Research · Data Visualization · Statistical Analysis · Data Storytelling

Education

Sept 2021 - May 2023 The University of New South Wales

M.S. Health Data Science

WAM: 88% (Dissertation: 91%)

Dissertation: Changes in atherosclerotic cardiovascular disease risk over time among people living with HIV

Mar 2015 - Mar 2019 The University of Sydney

B.S. Pharmacy

Employment History

Research-focused Positions

Mar 2023 - Dec 2023; Data Scientist / Research Officer, Sydney Pharmacy School, The University of Sydney
April 2025 - Present

Lead analyst on national studies using large-scale linked health data to map multimorbidity clusters and high-risk prescribing in older Australians. Experienced in study design, statistical analysis, cohort development, and translating complex data into actionable research outputs.

- **Lead analyst and first author, Multimorbidity clusters among males and females aged 65 years and over in Australia with at least one pharmaceutical or medical service claim: a nationwide cross-sectional data linkage study.** Responsibilities included conducting literature reviews, contributing to study design, performing statistical analyses (including network cluster analysis), developing advanced data visualisations, and drafting the first-author manuscript. Outputs include first-author manuscript accepted for publication in the Medical Journal of Australia and contributing to conference abstracts and presentations.
- **Lead analyst, Multi-Agency Data Integration Project (MADIP)** investigating high-risk prescribing patterns in older adults with dementia using linked PBS and Census data. Responsibilities included literature review, study design, descriptive analyses, and logistic regression modelling. Outputs included contributing to manuscripts, conference abstracts, and presentations.
- **Lead analyst, Developed a comprehensive national dementia cohort** integrating hospital admissions, ED presentations, PBS, MBS, aged care, and death registrations

July 2023 - Mar 2025 **Data Scientist, School of Population Health, The University of New South Wales**

Analyst in managing and analyzing large, linked health and social datasets, generating insights through statistical analysis and visualizations.

- **Analyst, Aboriginal Community Controlled Ear Health Support System (ACCESS) project.** Responsibilities included conducting literature reviews, drafting data management and analysis plans with research teams, preparing analysis-ready datasets from linked sources, and performing statistical analyses including descriptive statistics and visualizations to examine ear health patterns among Aboriginal children over time under supervision.
- **Analyst, Evaluation of Sustaining NSW Families project.** Responsibilities included conducting literature reviews, drafting and implementing data management and analysis plans for linked datasets, and performing analyses using target trial emulation with inverse probability of treatment weighting under supervision.
- **Assisted in building a large-scale linked data platform integrating multiple datasets for NSW Child E-Cohort Project.**

Research Interests

My research interests involve in ageing, multimorbidity clusters and trajectories, and high-risk prescribing, with a particular focus on developing and applying advanced statistical and analytical methods to large-scale linked health administrative data.

Publications

[Google Scholar Profile](#)

A. Peer-reviewed Journal Articles

1. **Chen, W.**, Lu, C. Y., Hilmer, S., Gibson, A., Tan., E. C. K (2026) "Multimorbidity clusters among males and females aged 65 years and over in Australia with at least one pharmaceutical or medical service claim: a nationwide cross-sectional data linkage study" *Manuscript accepted for publication in the Medical Journal of Australia on 20 May 2026*.
2. **Chen, W.**, Petoumenos, K., Somia, A., Edmiston, N., Chaiwarith, R., Woolley, I., Ross, J., Pujari, S., & Boettiger, D. C (2024) "Changes in atherosclerotic cardiovascular disease risk over time among people living with HIV" *Journal of Antimicrobial Chemotherapy*. 79(4), 897–902. DOI: [10.1093/jac/dkac049](#)
3. Lau, E. C. Y., **Chen, W.**, Hilmer, S. N., Jeon, Y.-H., Lu, C. Y., Fujita, K., Wang & Tan, E. C. K. (2025) "High-risk geriatric prescribing in older people with and without dementia in Australia: a nationwide cross-sectional study" *Journal of the American Medical Directors Association*. 27(2):106006. DOI: [10.1016/j.jamda.2025.106006](#)
4. Le, H. T., Lau, E. C. Y., **Chen, W.**, Lu, C. Y., Nguyen, T. A., Low, L.-F., Hilmer, S. N., Jeon, Y.-H., & Tan, E. C. K. (2025) "Prevalence and Risk Factors for Psychotropic Medication Use in Older Adults in Australia: A Nationwide Data Linkage Study" *Drugs & Aging*. 42(7):755–769. DOI: [10.1007/s40266-025-01220-6](#)
5. Huynh, A. L. H., Wang, Y., Ma, L., Low, Y. C. L., **Chen, W.**, Fowler, C., Tan, E. C. K., Masters, C. L., Jin, L., Pan, Y., & the AIBL Research Group. (2024) "A comparison of an Australian observational longitudinal Alzheimer's Disease cohort to Community-Based Australian data." *Journal of Alzheimer S Disease*. 1–13. DOI: [10.3233/JAD-240241](#)

6. Lau, E. C. Y., **Chen, W.**, Lu, C. Y., Hilmer, S. N., Jeon, Y.-H., & Tan, E. C. K. (2024) "Antidementia and psychotropic drug use in older people with dementia in Australia: a national data linkage study" *Journal of the American Medical Directors Association*. 25 (11), 105237. DOI: [10.1016/j.jamda.2024.105237](https://doi.org/10.1016/j.jamda.2024.105237)